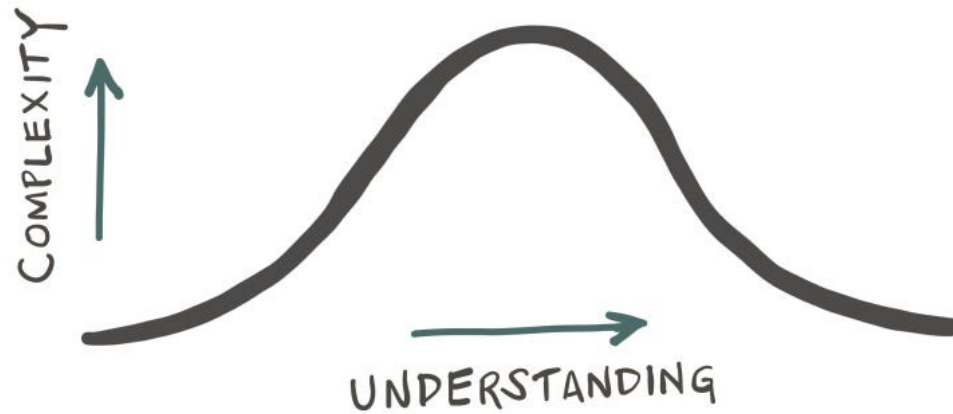




easier AI ?

timmm@ieee.org

ssbse'26

july 6, 2026

code: **pip install ezr**
data: **<http://tiny.cc/moot>**

Ssbse'16

[0]

An (Accidental) Exploration of Alternatives to Evolutionary Algorithms for SBSE

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Abstract. SBSE researchers often use an evolutionary algorithm to solve various software engineering problems. This paper explores an alternate approach of sampling. This approach is called SWAY (Sampling WAY) and finds the (near) optimal solutions to the problem by (i) creating a larger initial population and (ii) intelligently *sampling* the solution space to find the best subspace. Unlike evolutionary algorithms, SWAY does not use mutation or cross-over or multi-generational reasoning to find interesting subspaces but relies on the underlying dimensions of the solution space. Experiments with Software Engineering (SE) models shows that SWAY's performance improvement is competitive with standard MOEAs while, terminating over an order of magnitude faster.

Keywords: Search-based SE · Sampling · Evolutionary algorithms



Credits



- 2019:
 - **Vivek Nair** (Meta)
 - **Amritanshu Agrawal** (Snap)
 - **Jianfeng Chen** (Meta)
- 2025: **Andre Lustosa** (Redhat)
- 2027:
 - **Kishan Ganguly**
 - BINGO (FSE'26)
 - **Amirali Rayegon**
 - Explanation (JSS'26)
- 2028: **Srinath Srinivasan**
 - Neurosymbolic (arxiv)

why easier AI?

how to do easier AI?

why not more of easier AI?

why easier AI?

how to do easier AI?

why not more of easier AI?



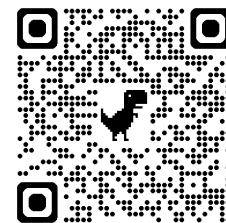
Lehman's 2nd law [1][2]

- “As a system evolves, its complexity **increases** unless work is done to maintain or reduce it.”
- Q: why / how can we **reduce** the complexity of AI?
- If anyone can do it for \$1:
 - Can you do it for a cent?
- Modern debloat menu:
 - distillation, quantization, small-language-models
 - You know we can go (much) smaller, still, right?



Q: why easier AI?

A: economics [3]

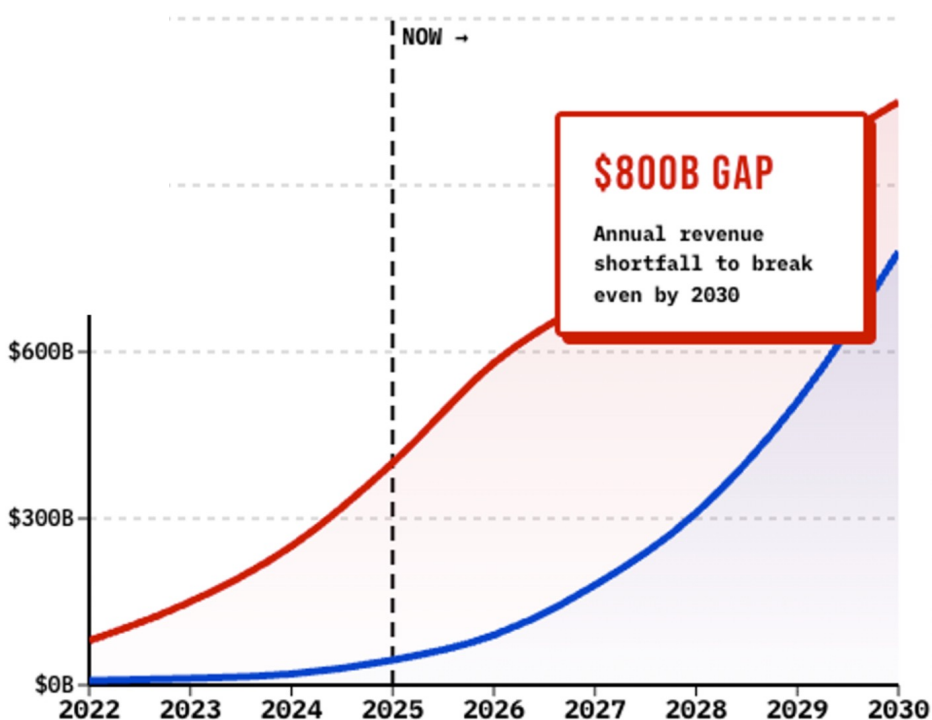
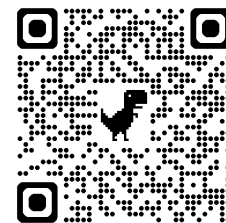


- Niklaus Wirth:
 - “Software gets **slower** faster than hardware gets faster”
- Even if \$/token *falls*, Jevon’s Paradox applies:
 - tasks/day and tokens/task will increase
 - +50% (or more) LLM cost (market **correction**)
 - Historically: subsidize (for market penetration), then cost: +100% (Uber)
 - *10, *100 more token burn (2023 → 2026)
- Very very soon, we will be asking
 - “how **to mix and match** simpler and complex AI?”



Q: why easier AI ?

A: cause we can't afford the alternative [4]



- Since this 2025 forecast:
 - 2026 capex ~\$725B,
 - 2027 projected >\$1T
 - I.e. gap widening

Q: why easier AI?

A: easier to trust; easier to watch and understand



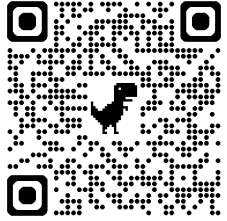
My Tesla Was Driving Itself Perfectly —Until It Crashed

The danger of almost-perfect tech

By Raffi Krikorian



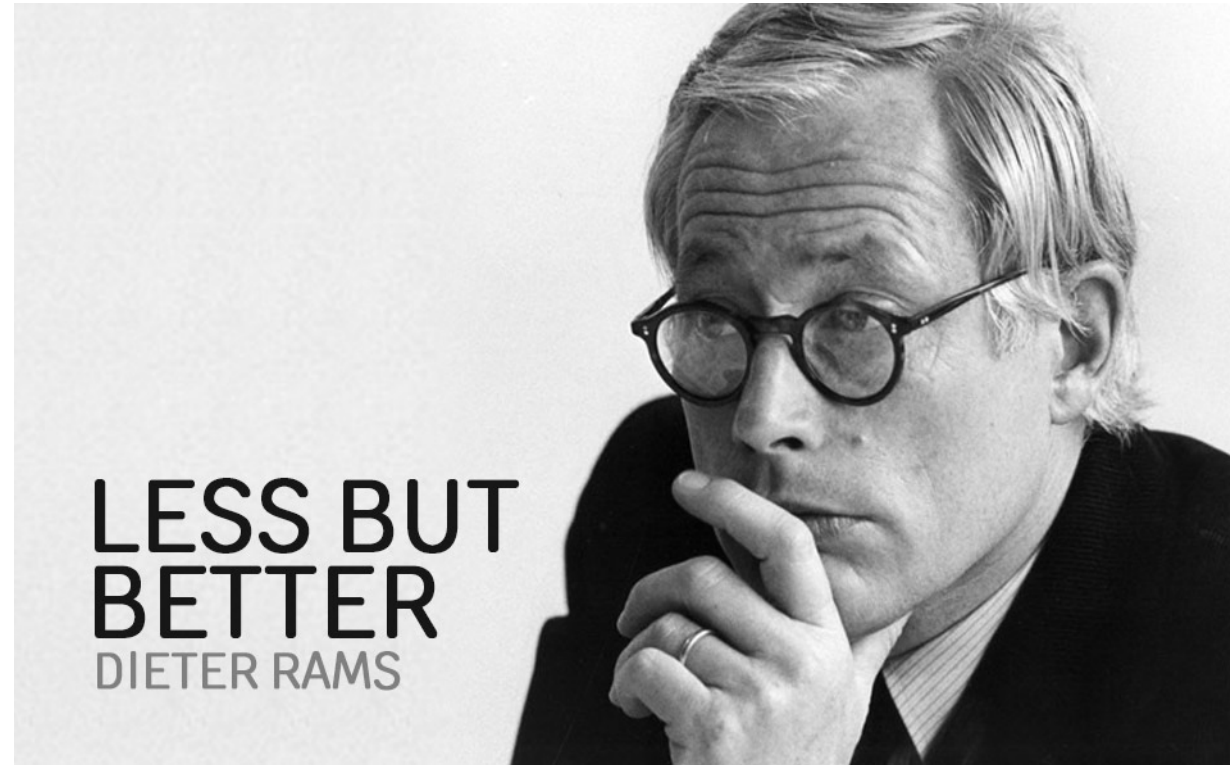
THE **ATLANTIC**
MARCH 17, 2026



Q: why easier AI?

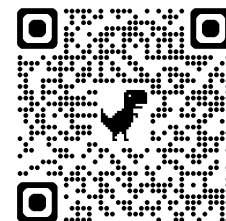
A: cause some domains demand it [5]

- Green AI
- global **south** (countries without e.g. American-scale infrastructure)
- resource-constraint teaching
 - community colleges,
 - K-12 CS,
 - MSIs,
 - these using **personal** hardware
- privacy-constrained industrial dev
 - **defense**,
 - regulated healthcare,
 - financial services,
 - law firms)
- air-gapped settings;
- **edge** or on-device inference

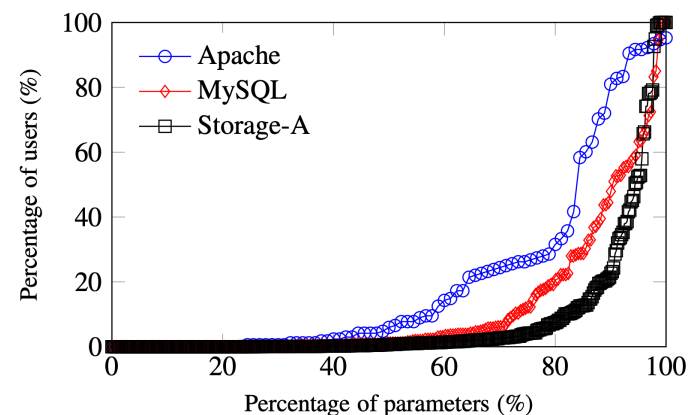
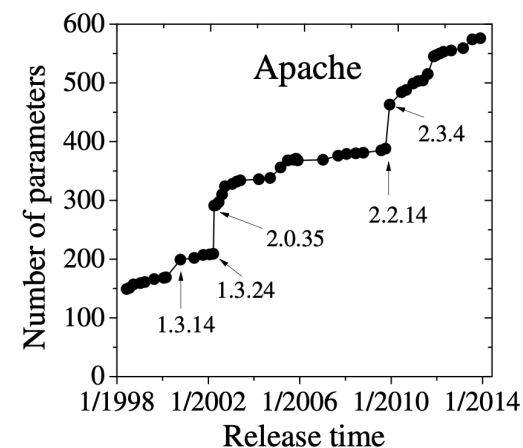


Q: why easier AI?

A: software engineering



- **security:**
 - modern software = assembly of pre-made parts (> 80%)
 - more reuse == more exposure
 - e.g. left-pad; MOVEit (\$12B), Falcon (\$5B), ByBit (1.4B) \$24B (2025);
- **survivability:** e.g. SQLite does not use Berkeley DB.
 - Now used on Mars
- **maintainability:** more loc = more to maintain
- **reliability:** more loc = more that can go wrong
- **waste:** Most users cannot handle most options [6]



Q: why easier AI?

A: cause “big AI” needs too much data [7]

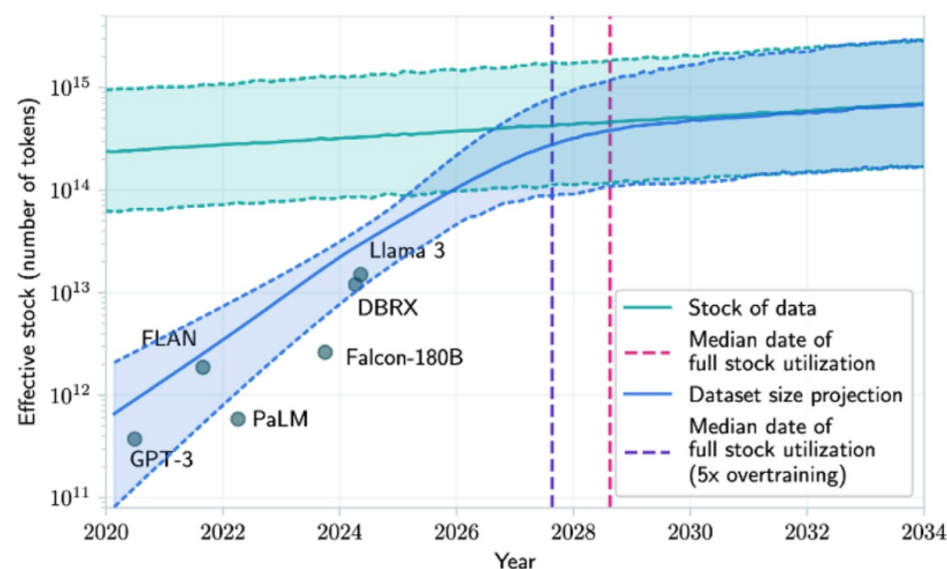
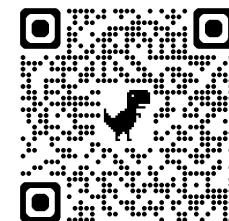
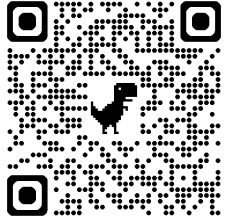


Fig. 4. Median results from the Villalobos et al. model (shown in green) estimate that by 2028, we will run out of new textual data needed to train bigger and better LLMs [75].



Q: why easier AI?

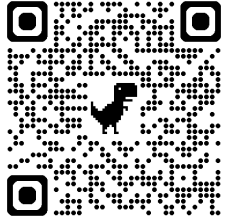
A: more AI not always better [8][9][10][11-14]

- text mining predicting effort within **agile** projects,
 - (SVM and TFIDF) better predictions than deep learners
 - Runs 100 times faster
- text mining, programmer discussion boards
 - simpler pre-clustering the data with K-Means comparable with deep learning (500-1000 times **faster**)
- parametric statistical methods > GAN-based
- as of a 2025 lit review, many examples traditional ML methods > LLMs
 - achieve **comparable** or superior performance with significantly less computational overhead
- tree-based models (Random Forest, XGBoost)
 - consistently **outperform** deep neural networks on tabular datasets,
 - especially those with more than 10,000 rows

why easier AI?

how to do easier AI?

why not more of easier AI?



Q: how to do easier AI?

A: so many ways [15-28]

- 130 A.D., **Ptolemy** “good principle to explain the phenomena by the simplest hypothesis possible”
- 1300: William of Ockham: “entities must not be multiplied beyond necessity”
- 1902, **PCA**: reduce data to a few principal components
- 1960s, Narrows: guide search via a few key variables
- 1974, **Prototypes**: speed up k-NN by reducing rows to a few exemplars
- 1984, JL lemma: random projection a few dimensions can preserve pairwise distances (within some error)
- 1986, **ATMS**: only focus diagnosis on core assumptions
- 1994, ISAMP: a few restarts can explore large problem spaces
- 1996, Sparse coding: learn efficient, sparse representations from data which inspired dictionary learning and **sparse autoencoders**
- 1997, Feature selection: ignore up to 80\% of features
- 2002, **Backdoors**: if we first set a few variables, that cuts exponential time to polynomial
- 2005, Semi-supervised learning: data can be approximated on a much lower-dimensional manifold
- 2009, **Active learning**: only use most informative rows
- 2003–2021, SE “keys”: a few parameters govern many SE models
- 2010+, **Surrogates**: first, build small models to label the rest of the data
- 2020s, Distillation: compress large LLM models with little performance loss

Q: how to do easier AI (with active learning)
A: use current model to decide what's next [29]



- E.g. **SMAC3**
 - ensemble of decision trees
 - Seek unlabeled examples where
 - ensemble most disagrees (a.k.a. explore)
 - **ensemble** most agree (a.k.a. exploit)
 - adapt =
Slowly shift explore to exploit
- Cost of SMAC3:
 - when new data arrives
 - **rebuilding** the ensemble
- EZR: **pip install ezr**
- “ensemble” = 2 **class** Bayes classifier
 - $|Best|, |Rest| = \sqrt{N}, N - \sqrt{N}$,
 - **next** =
 - closest to *Best*
 - furthest from *Rest*
 - update:
 - add next to *Best*,
 - resort *Best*,
 - **pop** worst *Best* to *Rest*
 - Very fast
 - never sort *Rest*
 - Updates to *Best* and *Rest* are fast.

Summarize
symbols

```
Sym = dict
def Num(n=0, mu=0, m2=0): return (n, mu, m2)
```

Summarize
numbers

```
def count(sym, v, inc=1):
    if (c := sym.get(v, 0) + inc) > 0: sym[v] = c
    else: sym.pop(v, None)
    return sym
```

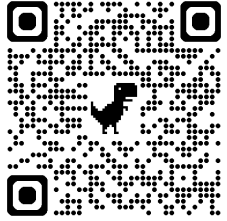
Add or subtract
symbols

```
def welford(num, v, inc=1):
    n, mu, m2 = num
    if (n := n + inc) ≤ 0: return Num()
    d = v - mu; mu += inc * d / n
    return (n, mu, m2 + inc * d * (v - mu))
```

Add or subtract
numbers

```
def mix(i, j, inc=1):
    if is_sym(i):
        return {k: i.get(k, 0) + inc * j.get(k, 0) for k in i | j}
    (ni, mui, m2i), (nj, muj, m2j) = i, j
    n = ni + inc * nj
    if n ≤ 0: return Num()
    d = muj - mui
    mu = (ni * mui + inc * nj * muj) / n
    m2 = m2i + inc * m2j + inc * d * d * ni * nj / n
    return Num(n, mu, max(0, m2))
```

Mix or separate
two columns



Q: how to test EZR?

A: read lots of papers, collect their data [30]

- Recent SBSE **papers**: ICSE, FSE, TSE, IST, EMSE, TOSEM, ASE
- 127 data sets
- 3-1044 independent **x attributes**;
- 1 to 8 dependent **y attributes** to be minimized or maximized;
- 93 to 100,000 rows.
- <http://tiny.cc/moot>

#	Category	Focus	File Name(s)	Here, optimizers struggle with issues like...	#y	#x	Rows	Refs
Software Systems & Configuration & Tuning								
24	Soft. Config	PLE	SS-[A-X]	Minimize footprint/memory vs. maximizing throughput	2-3	3-88	197-86k	[1]
12	PromiseTune	Perf.	7z, LLVM, BDBC	Minimize execution time and energy consumption	1	6-35	864-166k	[2]
1	Cloud compute	Tuning	Apache_AllMeas	Balance server response vs. CPU/RAM load patterns	1	9	192	[3]
1	Cloud compute	Tuning	SQL_AllMeas	Minimize latency/IO vs. maximizing throughput	1	39	4,654	[4]
1	Cloud compute	Tuning	X264_AllMeas	Optimize encoding parameters for PSNR/SSIM quality	1	16	1,153	[1]
2	Testing	Testing	test120, 600	Maximize coverage while minimizing execution time	1	9	5,161	-
Project Management & Process Modeling								
35	Proj. Health	Health	Health-Closed	Optimize PR rates and minimize developer churn	2-3	5	10,001	[5]
3	Scrum	Agile	Scrum[1k-100k]	Maximize velocity within sprint constraints	3	124	1k-100k	[6]
8	Feature Mod.	Config	FFM, FM-*	Optimize Clause/Constraint ratio in large spaces	3	128-1k	10,001	[1]
1	nasa93dem	Cost	nasa93dem	Minimize effort (person-months) vs. quality	3	26	93	[5]
1	COC1000	Risk	coc1000	Minimize risks/schedule slip vs. analyst expertise	5	20	1,001	[7]
4	POM3	Process	pom3[a-d]	Balance project idle rates vs. completion costs	3	9	501-20k	[6]
4	XOMO	Defects	xomo_[flt,grd]	Minimize defect density vs. total project effort	4	27	10,001	[7]
Interdisciplinary & Behavioral Benchmarks								
4	Behavioral	Behavior	all_players, etc	Maximize user retention and predict cohort churn	1-3	26-55	82-17k	[8]
4	Financial	Finance	BankChurners	Minimize credit risk vs. optimal pricing models	2-5	19-77	1k-20k	[9]
3	Health	Health	COVID19, Life	Maximize accuracy/life vs. readmission likelihood	1-3	20-64	2k-25k	[10]
2	RL	Control	A2C_[Acr,Crt]	Maximize reward signals vs. steps to convergence	3-4	9-11	224-318	-
5	Sales	Market	Marketing, socks	Maximize ROI vs. minimizing forecasting error	1-8	20-31	247-2k	[11]
3	Misc.	General	auto93, Car	Optimize multivariate trade-offs (e.g. MPG vs. HP)	2-5	5-38	205-1k	[1]

configuration, performance tuning, product lines, project health, defect prediction, testing, cost estimation, cross-domain generalization, and text mining



Q: how to score EZR?

A: win = 1 - normalize(regret) [31]

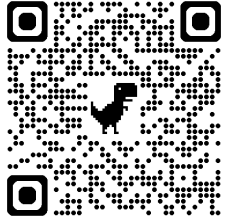
- Distance to **heaven** (aggregate n goals to one)

$$d2h(y) = \sqrt{\sum_{i=1}^k y_i^2 / \sqrt{k}},$$

- Regret = (**d2h** - lo) / (mu - lo)
- Win = int(100 * (1 - regret))
 - Win=0 → optimizer just finds mu
 - Failure**
 - Win=100 → found reference optimum
 - Success
- Build regression tree on **labeled** eggs
 - Splitting rows to minimize σ of win
 - Reports what takes you towards good or ba

win	n	Lbs-	Acc+	Mpg+	
8	43	2334.2	16.3	27.4	
41	26	2038.7	17.1	30.0	Volume <= 111
71	10	1871.7	18.1	32.0	Volume <= 83
✓ 85	4	1831.7	19.0	32.5	Model <= 73
62	6	1898.3	17.4	31.7	Model > 73
22	16	2143.0	16.5	28.7	Volume > 83
36	7	2098.1	18.0	28.6	Model <= 73
12	9	2177.9	15.3	28.9	Model > 73
26	4	2210.0	15.7	30.0	Volume > 97
0	5	2152.2	15.0	28.0	Volume <= 97
-41	17	2786.1	15.1	23.5	Volume > 111
-9	5	2279.8	14.6	28.0	Volume <= 116
-54	12	2997.1	15.3	21.7	Volume > 116
-44	8	2921.1	15.8	22.5	Model > 73
-38	4	2563.0	14.0	25.0	Clnhrs <= 4
-50	4	3279.2	17.4	20.0	Clnhrs > 4
✗ -75	4	3149.0	14.4	20.0	Model <= 73

configuration, performance tuning, product lines, project health, defect prediction,
testing, cost estimation, cross-domain generalization, and text mining

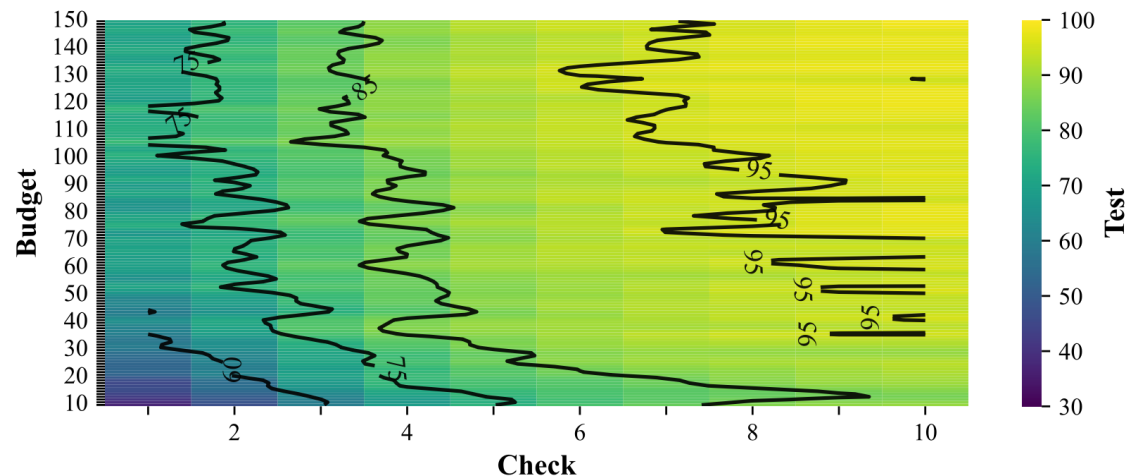


Q: how well does EZR generalize?

A: sort holdout via tree, check first few [31]

$\text{Win} = \text{int}(100 * (1 - \text{regret}))$

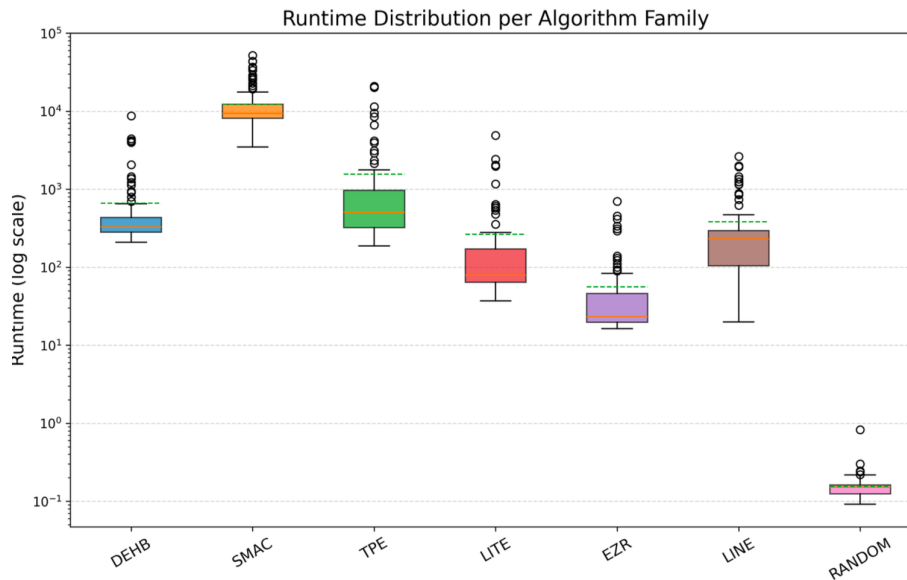
- 100,000 times
 - pick any MOOT data set
 - Train:
 - Active learning, labeling y rows with a **limited** budget.
 - Build tree from labels
 - Test:
 - Use tree to sort holdout
 - **Check** first few
- Active learning, very effective
 - Label 50 rows
 - Check 5
 - **Achieve** 85% of best



This entire 100,000-experiment study: one laptop, 2 hours, no cluster, no cloud bill.

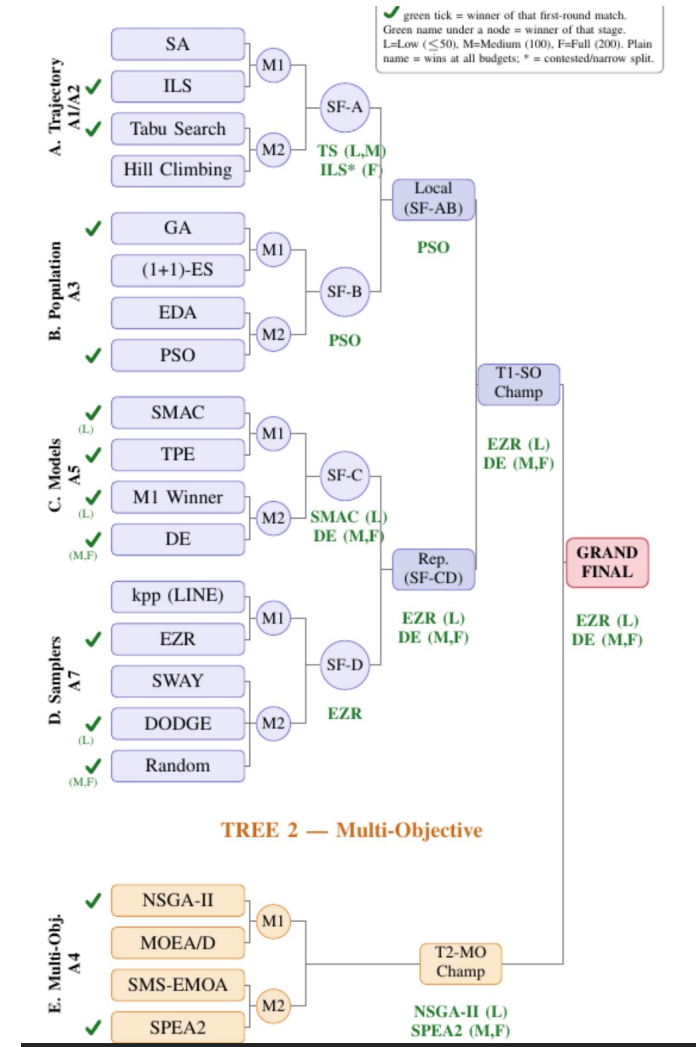
Q: how fast and effective
are EZR's trees?

A: very and very [32]



ms

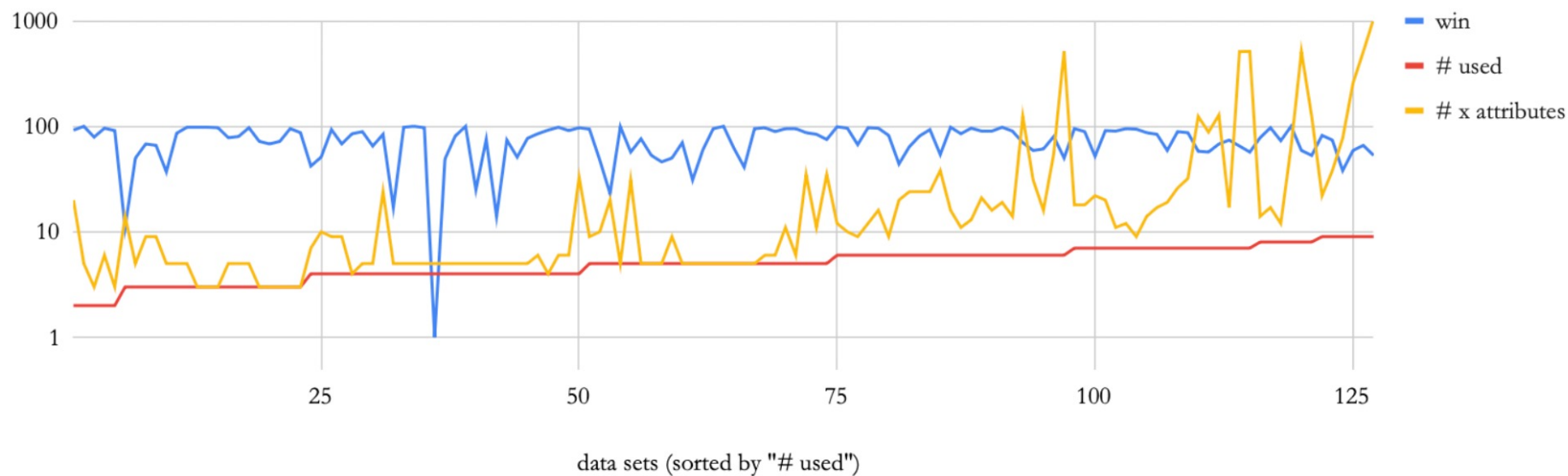
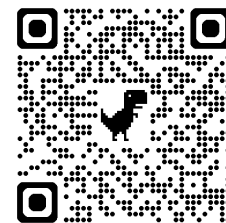
500x less CPU = 500x less energy. every run.

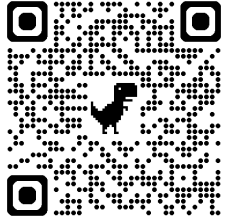


e.g. NSGA-II needs 1000 samples for what EZR reaches in 50

Q: how big are EZR's models?

A: surprisingly small **[33]**



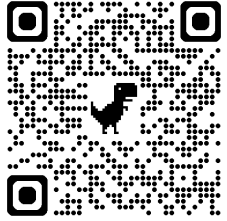


Q: so is EZR useful for explanation?

A: EZR's trees competitive with SOTA [33]

- Build trees using N **attributes**
- N= #attributes used in EZR's tree
- Using other tools:
 - select **top** N attributes
- White = tied best
- Blue = within **top** 90%
- Yellow = within top 75%
- EZR goes a long way
 - E.g. last row
 - For a **3 goal** task
 - EZR uses 7 / 1044 attributes
 - EZR uses 150 rows
 - Arguably competitive

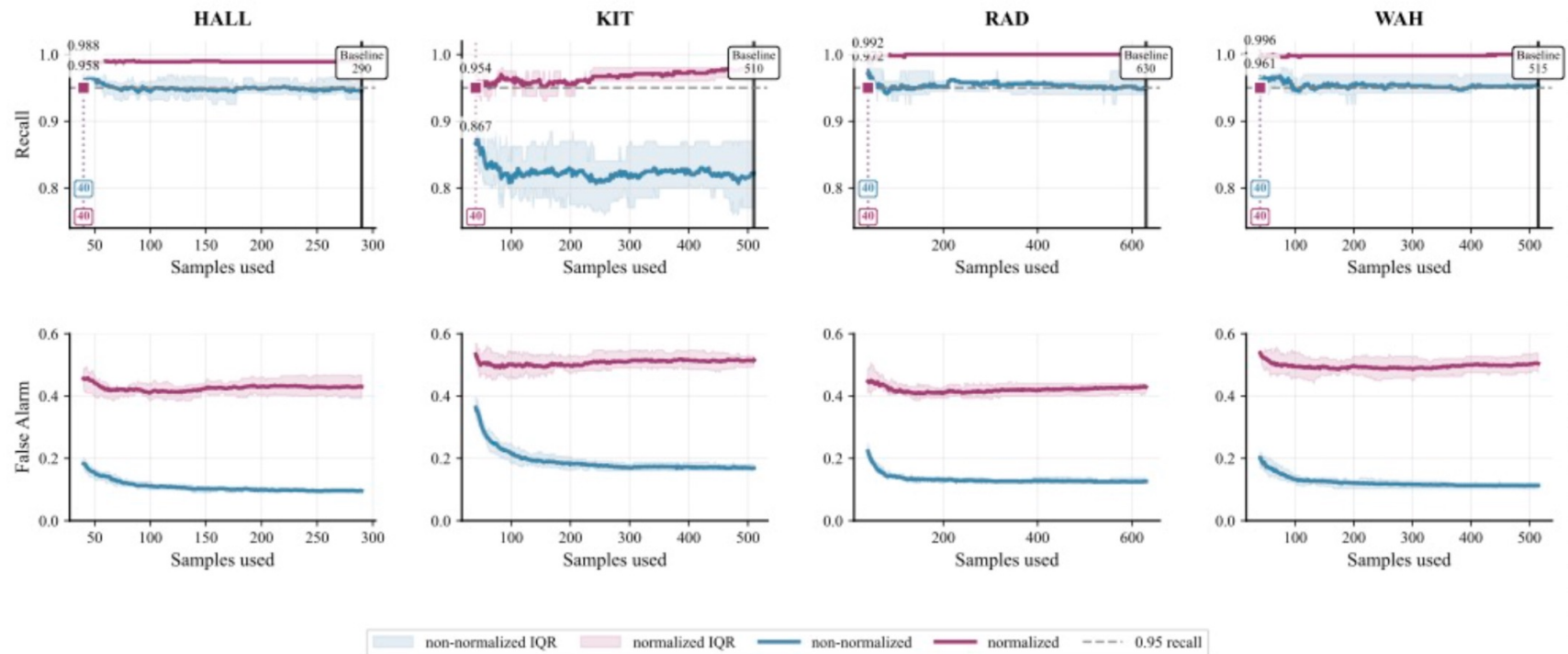
RLF	SHAP	EZR	anova	all	data	x*	x	y	budget	rows
95	100	100	100	100	HSMGP	6	14	1	150	3457
87	81	82	82	85	SS-R	7	14	2	150	3009
94	95	63	57	96	SS-V	4	16	2	150	6841
74	74	59	74	99	SS-W	7	16	2	150	65537
100	100	100	100	100	X264-AM	6	16	1	150	1153
100	100	100	99	100	SS-M	7	17	3	150	865
81	81	81	81	81	SS-N	8	17	2	150	53663
59	62	53	52	58	coc1000	9	20	5	150	1001
95	96	96	93	100	SS-U	7	21	2	150	4609
98	98	93	98	98	xomo_flight	7	27	4	150	10001
92	92	90	90	92	xomo_grnd	7	27	4	150	10001
94	95	95	95	95	xomo_osp	7	27	4	150	10001
87	88	82	82	88	xomo_osp2	8	27	4	150	10001
77	75	71	64	85	SQL-AM	10	39	1	150	4654
87	91	92	74	94	billing10k	7	88	3	150	10001
75	75	81	75	97	Scrum100k	7	124	3	150	100001
82	88	92	88	97	Scrum10k	8	124	3	150	10001
83	89	86	83	89	Scrum1k	7	124	3	150	1001
81	99	93	81	99	FFM-125	9	128	3	150	10001
81	97	89	97	97	FFM-250	8	256	3	150	10001
95	93	93	95	95	FFM-500	8	510	3	150	10001
81	96	87	96	96	FM-500-1	8	511	3	150	10001
86	91	93	91	97	FM-500-2	7	511	3	150	10001
95	95	95	87	95	FM-500-3	9	513	3	150	10001
97	97	93	96	97	FM-500-4	9	517	3	150	10001
82	94	94	71	100	FFM-1000	7	1044	3	150	10001



Q: what does EZR say about LLMs?

A: LLMs rarely tested against simpler tool [35]

- Search google scholar for **relevant** research



why easier AI?

how to do easier AI?

why not more of easier AI?

Q: if so good, why not more easier AI?

A: (weak) empirical standards [35] [36]

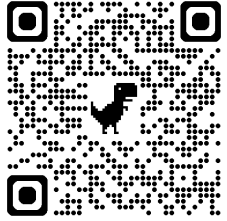


- In a survey of 229 SE papers on **LLMs in SE**
 - Only 5% compared to simpler approaches
- **Methodological** error
 - Other methods can produce results that are better and/or **faster**
 - (See above)
- My career:
 - Step 1: compare "it" vs a **stupid** straw man
 - Step 2: throw "it" away, use the straw man



Q: if so good, why not more easier AI?

A: cause prompts/ loops/ LLM-wikis are cool [37]



The screenshot shows a terminal window with a dark background. At the top, there's a title bar with three colored circles (red, yellow, green) and the text 'make sh'. The main area contains Python code for parsing CSV files and a function 'say'. The code is as follows:

```
[nothing] 0:htop*
15 if (s[1:] if s[:1] == "-" else s).isdigit(): return int(s)
16 try: return float(s)
17 except ValueError: return s=="True" or (s!="False" and s)
18
19 def settings(s):
20     "Parse every var=val in a string into an o (vals coerced)."
21     pairs = re.findall(r"(\w+)=(\S+)", s)
22     return o(**{k: thing(v) for k, v in pairs})
23
24 def csv(file, clean=lambda s: s.partition("#")[0].split(",,")):
25     "Yield typed rows from a CSV file ('#' starts a comment)."
26     with open(file, encoding="utf-8") as f:
27         for line in f:
28             row = [x.strip() for x in clean(line)] # strip once, here
29             if any(row): # skip blank/comment lines
30                 yield tuple(thing(x) for x in row) # hashable: rows key caches
31
32 def say(x, dec=2):
33     "Pretty string: whole floats as ints, else `dec` places."
34     if isa(x, float):
35         return str(int(x)) if x == int(x) else f"{x:.{dec}f}"
36     if isa(x, o): x = vars(x) # unwrap a namespace
37     if isa(x, dict): return "{" + ", ".join(f"{k}: {say(v,dec)}"
38                                     for k, v in sorted(x.items())) + "}"

~/gits/aiez/nuff/nuff.py 24,30 3%
```

Below the code, there's a system status bar showing memory usage (17.6G/24.0G) and swap usage (1.40G/3.00G). At the bottom, there's a footer with function key shortcuts: F1Help, F2Setup, F3Search, F4Filter, F5Tree, F6SortBy, F7Nice, F8Nice, F9Kill, F10Quit.

On the right side of the terminal, there's a sidebar with the following information:

- timms-MacBook-Air.loc 20:27 01-Jul-26
- Claude Code v2.1.197
- Opus 4.8 · Claude Max
- ~/gits/timm/misc/26/ssbse
- 3 MCP servers need authentication · run /mcp
- Fable 5 is back.
- Until July 7, you can use up to 50% of your plan's weekly usage limit on Fable 5. If you hit your limit, you can continue on Fable 5 with usage credits. Fable 5 draws down usage faster than Opus 4.8. [Learn more](#)
- dir:~/gits/timm/misc/26/ssbse branch:maste...
- ← for agents

Q: if so good, why not more easier AI?
A: cause “big” sells



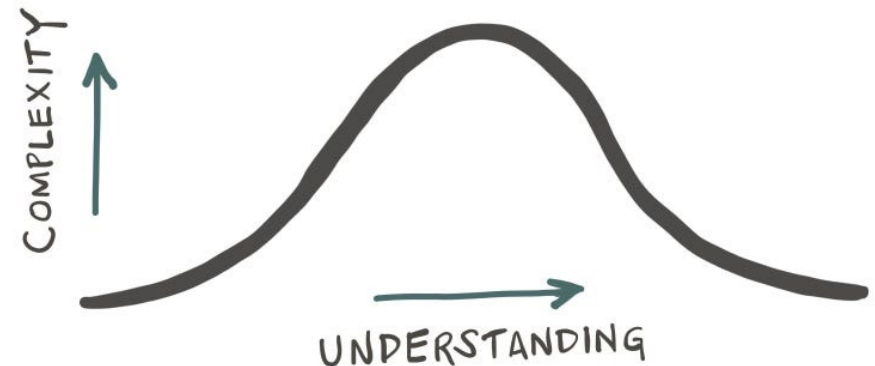
- Welcome to my data center
 - ribbon on the door the mayor can cut
 - **such** a spectacle!
- Welcome to **timmm inc.**
 - No special software to sell
 - cause you can build it yourself
 - No special hardware to sell
 - cause our algorithms are so fast, they don't **need** it
 - IPO ?
 - Low evaluation



Q: if so good, why not easier AI?
A: the simplicity paradox



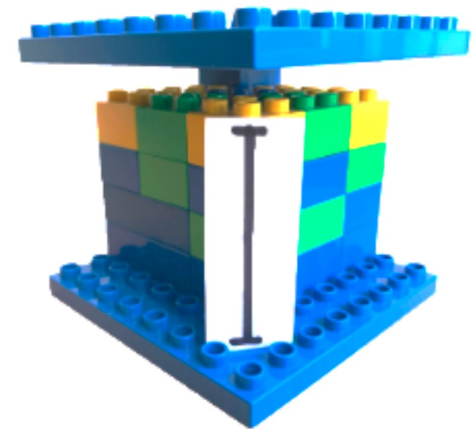
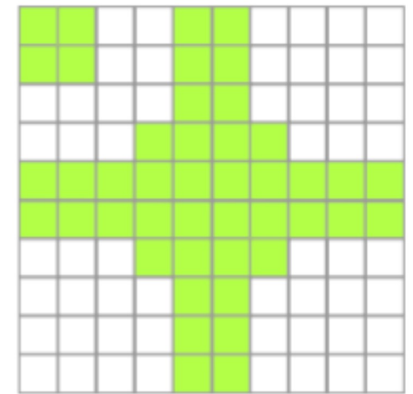
- Simplicity needs **experience**.
 - “Experience is the name every one gives to their **mistakes**.”
— *Lady Windermere’s Fan* (1892)
- Simplicity is complicated to certify
 - You have to build and/or **run** simpler “it” and complex “it”
 - Needs lot of CPU



Q: if so good, why not more easier AI?
A: human cognitive bias [38]



- Our simpler methods have not been **previously** explored
 - since other researchers **were not looking for them.**
- Why?
- Humans are biased to bloat
 - In **studies** of 100s of human subjects
 - Humans favor additive to subtractive changes
 - 1,155 to 297 (about 4:1)



Conclusion



Lehman's **2nd** law ^[1]

- “as a system evolves, its complexity **increases** unless work is done to maintain or reduce it”
- Q: why / how can we **reduce** the complexity of AI?
- A: we must / we can
- **do**: baseline the simple:
pip install ezr + tiny.cc/moot
- **review**:
 - ask “compared to what simpler?”
- **teach**:
 - easier AI first; scale only when simple fails



/ (question | comment)* /

code: *pip install ezr*
data: *<http://tiny.cc/moot>*

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